

BOONE, NC, USA

WMO: 722198

Lat: 36.200N

Lon: 81.650W

Elev: 910

StdP: 90.86

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.1	(c) -9.4	(d) -20.4	(e) 0.7	(f) -6.2	(g) -17.6	(h) 0.9	(i) -5.7	(j) 16.1	(k) -3.6	(l) 15.0	(m) -2.5	(n) 6.2	(o) 290	(p) 0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.9	19.9	27.7	19.6	27.0	19.4	22.4	25.3	21.5	24.8	20.9	24.7	3.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.7	23.4	20.9	17.4	23.1	19.8	16.3	22.1	70.5	25.6	67.7	24.8	64.7	24.5	36.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.2	9.1	-15.1	32.6	3.1	5.3	-17.3	36.3	-19.1	39.4	-20.8	42.4	-23.1	46.2
			-16.0	23.8	2.7	3.4	-17.9	26.3	-19.5	28.3	-21.1	30.2	-23.1	32.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	1.1	2.6	6.2	11.3	15.7	20.0	21.1	20.4	17.5	12.1	6.2	3.3
	DBStd	8.31	6.02	5.52	5.20	4.35	3.73	2.55	2.30	2.32	2.97	4.53	4.56	4.99
	HDD10.0	1048	281	212	141	36	5	0	0	0	1	31	129	214
	HDD18.3	2764	535	440	377	212	92	13	6	8	50	199	363	467
	CDD10.0	1596	4	5	22	75	183	301	343	323	224	94	16	5
	CDD18.3	270	0	0	0	1	12	64	91	73	24	5	0	0
	CDH23.3	1695	0	0	0	26	101	412	535	421	158	42	0	0
	CDH26.7	263	0	0	0	1	6	63	91	72	13	17	0	0
(14) Wind	WSAvg	2.8	4.2	4.2	3.7	3.3	2.5	2.2	1.9	1.6	1.5	2.5	3.0	3.5
(15) Precipitation	PrecAvg	1352	107	91	137	110	105	128	133	135	117	93	94	104
	PrecMax	1679	248	231	223	239	252	246	242	284	515	337	221	205
	PrecMin	977	32	32	44	25	34	24	73	25	8	3	19	27
	PrecStd	209	56	43	45	55	52	52	42	67	100	71	51	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.3	22.1	26.2	27.4	30.0	30.5	30.5	28.7	27.5	20.6	17.5
		MCWB	11.9	12.8	13.7	13.9	16.9	19.5	19.8	21.2	18.6	21.0	13.9	12.3
	2%	DB	14.1	15.6	19.0	23.7	26.1	28.4	28.8	28.7	27.0	23.0	17.8	15.4
		MCWB	10.7	11.2	11.5	13.5	16.8	19.1	20.0	20.6	19.1	16.3	11.9	11.8
	5%	DB	12.5	13.6	17.1	22.1	24.1	27.3	27.6	27.3	25.7	21.3	16.1	13.2
		MCWB	9.8	9.9	10.8	13.4	16.3	19.0	20.0	20.2	18.5	15.4	11.9	10.8
	10%	DB	10.2	11.6	14.7	19.7	22.8	26.1	26.3	26.0	23.7	19.0	14.0	11.9
		MCWB	7.9	8.1	9.5	12.4	16.0	18.5	20.0	19.9	18.1	15.1	10.4	9.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	14.7	14.6	17.0	19.6	21.9	23.3	23.3	23.4	22.8	17.2	15.4
		MCDB	15.2	17.1	19.1	21.3	23.9	25.2	25.9	26.7	23.9	25.5	17.4	16.2
	2%	WB	12.0	12.5	13.4	15.6	18.7	21.1	22.3	22.3	21.0	19.1	14.6	13.1
		MCDB	13.6	14.5	16.9	19.8	22.4	25.5	24.9	25.7	23.7	20.9	16.1	14.4
	5%	WB	10.3	10.7	12.2	14.7	17.9	20.4	21.4	21.5	20.0	17.6	13.1	11.6
		MCDB	12.3	12.9	15.1	18.9	21.5	25.1	24.7	24.9	23.2	19.3	14.5	12.6
	10%	WB	8.0	8.5	11.0	13.7	17.2	19.7	20.8	20.8	19.1	16.2	11.7	9.7
		MCDB	9.4	10.8	13.8	17.9	20.9	24.3	24.8	24.3	22.0	18.2	13.4	11.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.5	10.3	11.9	10.9	10.5	9.9	10.0	10.4	10.5	10.8	9.0
		MCDBR	11.2	12.2	13.5	16.0	13.1	11.8	11.4	11.8	13.2	13.6	13.3	11.4
	5% WB	MCWBR	7.9	8.3	7.4	7.7	5.9	5.0	4.8	5.2	6.2	7.6	8.2	8.2
		MCWBR	9.6	10.8	11.3	11.8	11.0	10.3	9.7	10.0	10.3	10.2	9.3	9.4
(40) Clear-Sky Solar Irradiance	taub		0.275	0.281	0.305	0.354	0.392	0.423	0.457	0.453	0.359	0.317	0.286	0.272
		taud	2.582	2.561	2.531	2.385	2.282	2.223	2.161	2.147	2.401	2.528	2.594	2.618
	Ebn at Noon		939	970	969	928	892	861	828	824	895	912	917	922
		Edh at Noon	74	86	97	118	133	141	150	149	109	88	73	68
(44) All-Sky Solar Radiation	RadAvg		2.41	3.04	4.14	5.19	5.70	6.10	5.68	5.29	4.53	3.65	2.75	2.09
	RadStd		0.17	0.35	0.27	0.40	0.43	0.42	0.40	0.26	0.41	0.43	0.25	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	+154	+94

Nomenclature: See separate page